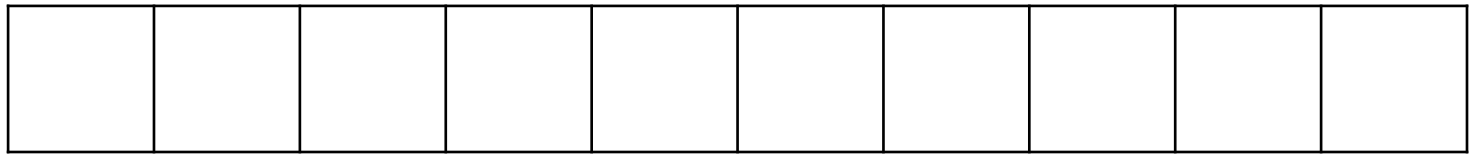
Cellular Automata

--A journey of rebirth and annihilation

Define a world



Status: Life, or Death?

0	0	0	0	0	1	0	0	0	0

Given a world

1st gen	0	0	0	0	0	1	0	0	0	1

1. Cells = 10
2. Initialize n cells = 1, others null(0)
3. Scan consequent cells by each 3
4. Update for next generation

	111	110	101	100	011	010	001	000
Rule 90	0	1	0	1	1	0	1	0

Determine via looking up rule30
Continuous border:
1, 0, 0, 0, 1, 0, 1, 0, 1, 0

1st gen	0	0	0	0	0	1	0	0	0	1
2nd gen	1	0	0	0	1	0	1	0	1	0

Rule the iteration

Think about neighbors: 000, 000, 000, 001, 010, 100, 000, 000

-- if ??? or ???:

to survive

--else:

to die

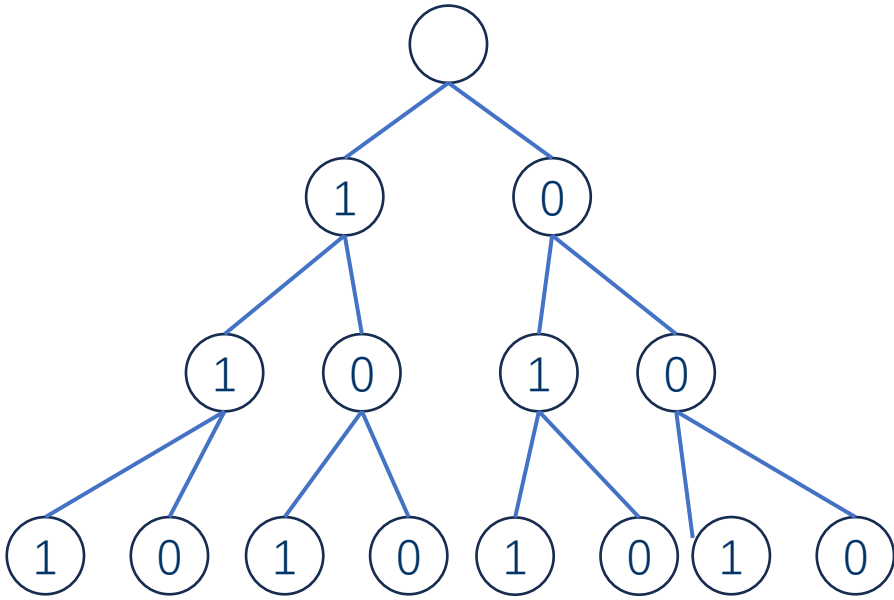
Current status:

0	0	0	0	0	1	0	0	0	0

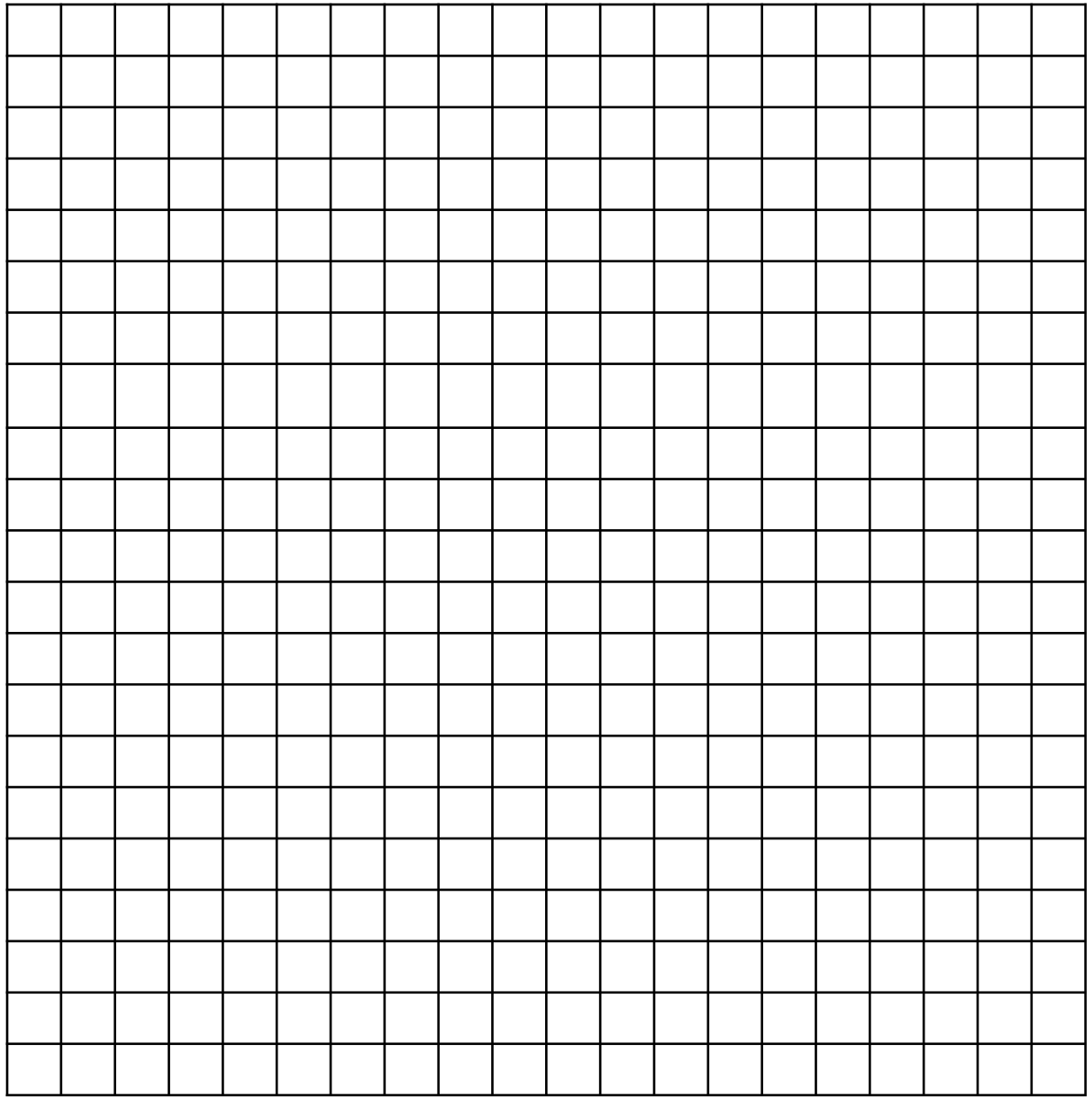
Iterate next generation:

?	?	?	?	?	?	?	?	?	?

Logical Determination by combinations of Left, Mid, Right



Dec	$2^7=128$	$2^6=64$	$2^5=32$	$2^4=16$	$2^3=8$	$2^2=4$	$2^1=2$	$2^0=1$
Bin	111	110	101	100	011	010	001	000
Rule**90**	0	1	0	1	1	0	1	0
Rule**30**	0	0	0	1	1	1	1	0



1. Initialize n cells
2. Count live neighbors
3. Update for next generation
4. Iteration 2~3

*Finish in advance if the maximum number of live neighbors of all cells is less than 2

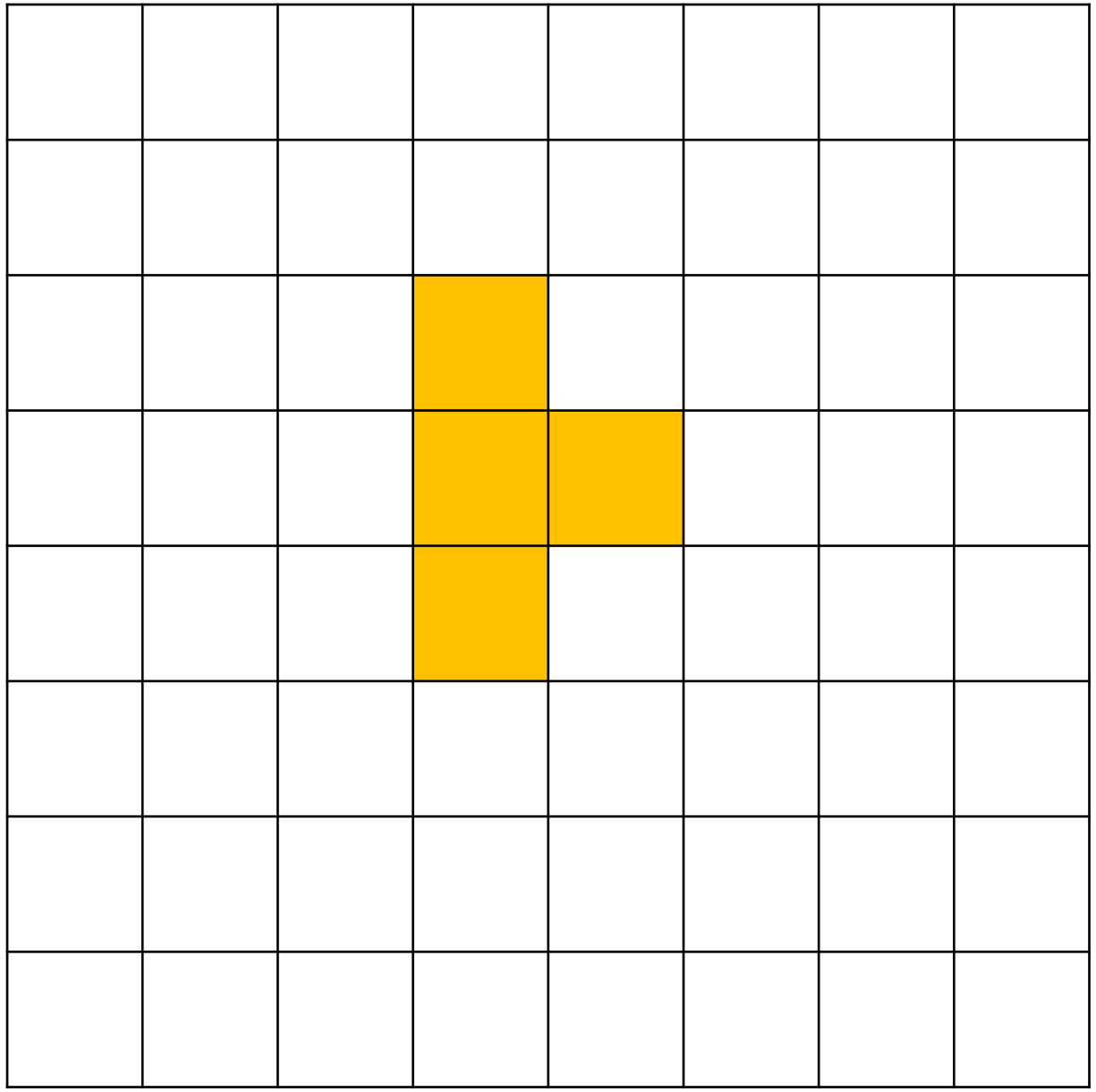
Grid: 8*8

Logical Determination

Survive: Live Neighbors = 2

Survive or Birth: Live Neighbors = 3

Die: Live Neighbor < 2 or >3

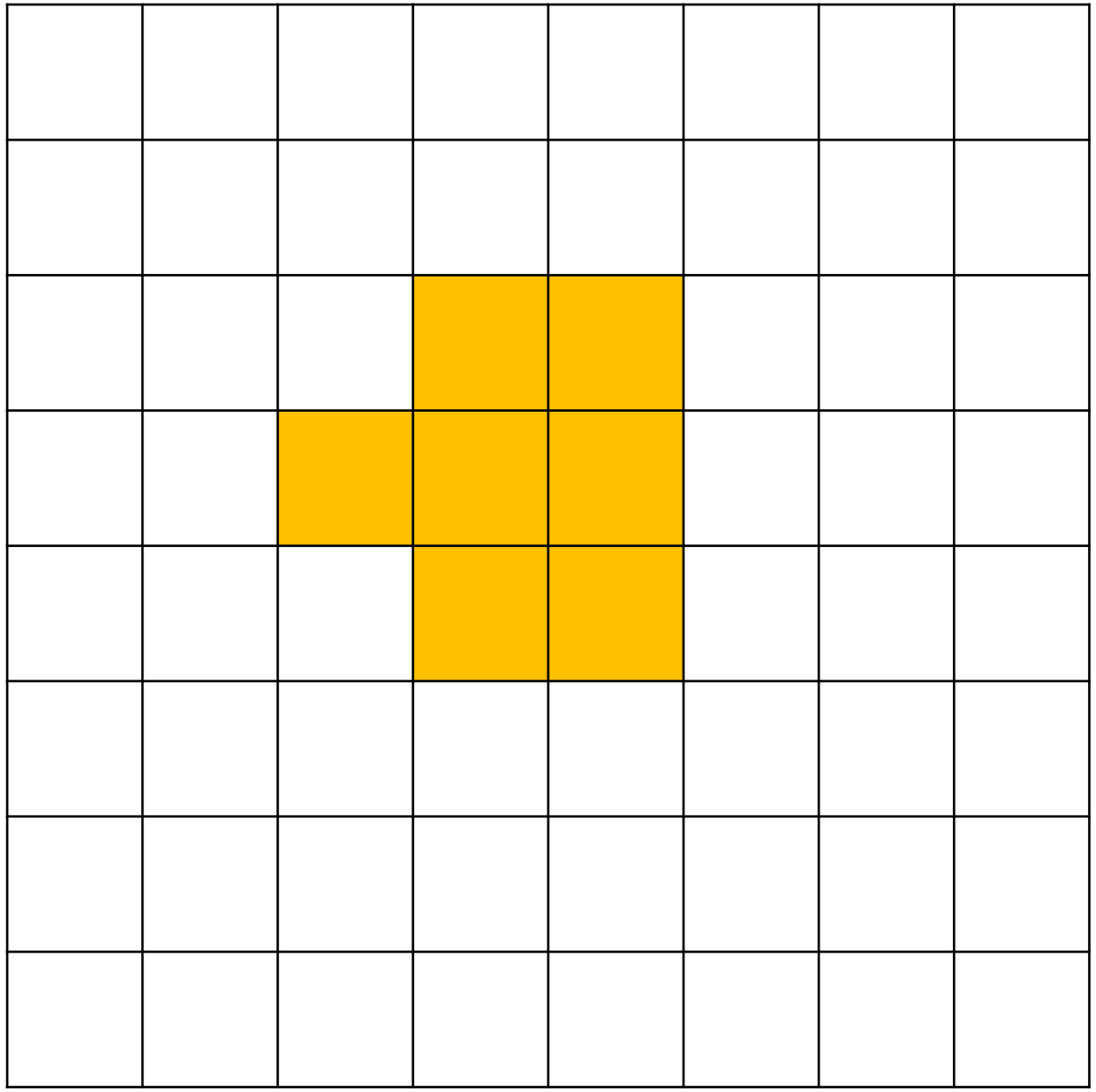


--Start:

1.Initialize 4 cells

		1	1	1			
		2	2	3	1		
		3	3	3	1		
		2	2	3	1		
		1	1	1			

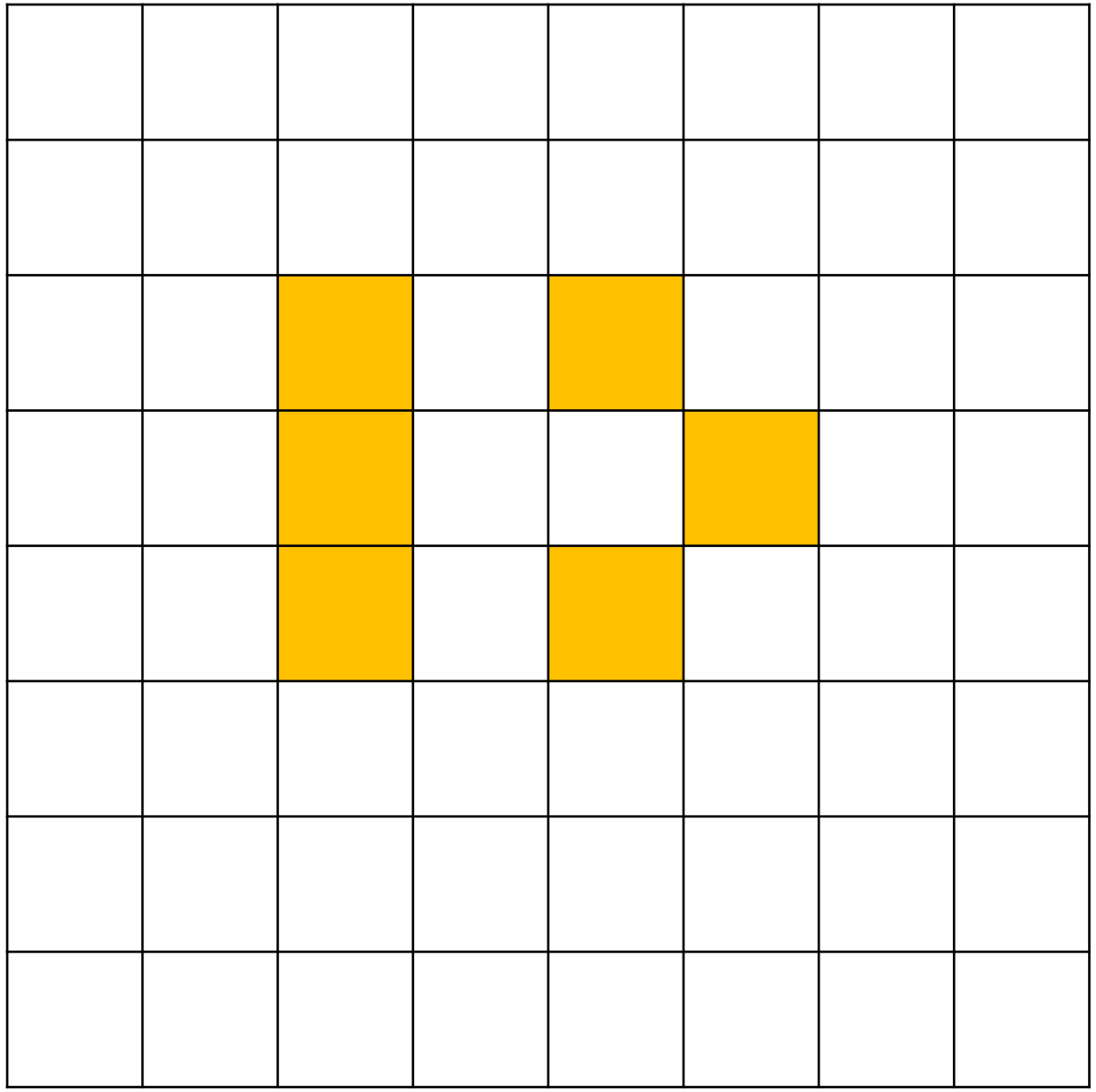
2. Count live neighbors



3. Update for next generation

		1	2	2	1		
	1	3	4	3	2		
	1	3	6	5	3		
	1	3	4	3	2		
		1	2	2	1		

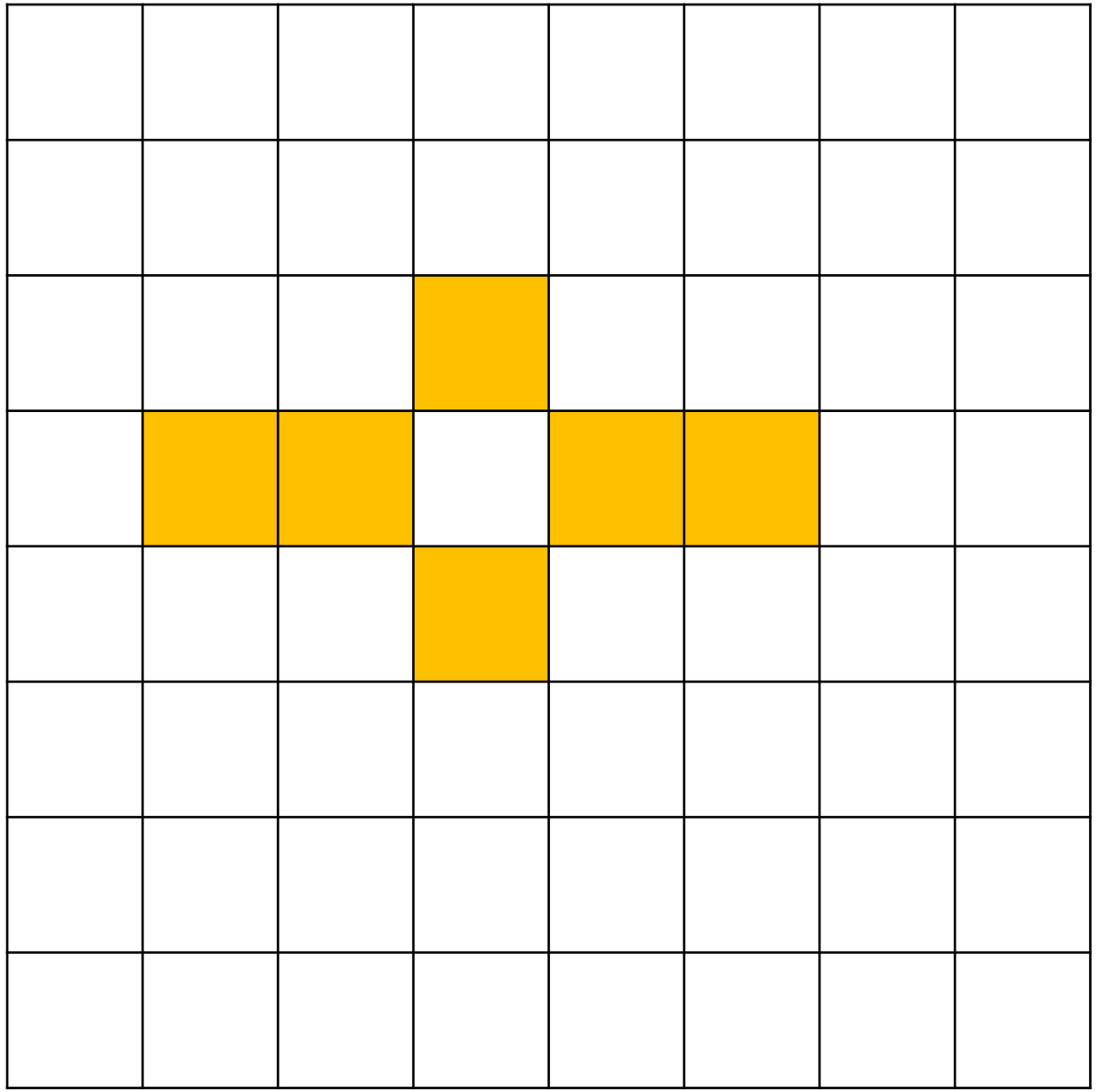
2. Count live neighbors



3. Update for next generation

	1	1	2	1	1		
	2	1	3	1	2	1	
	3	2	5	3	2	1	
	2	1	3	1	2	1	
	1	1	2	1	1		

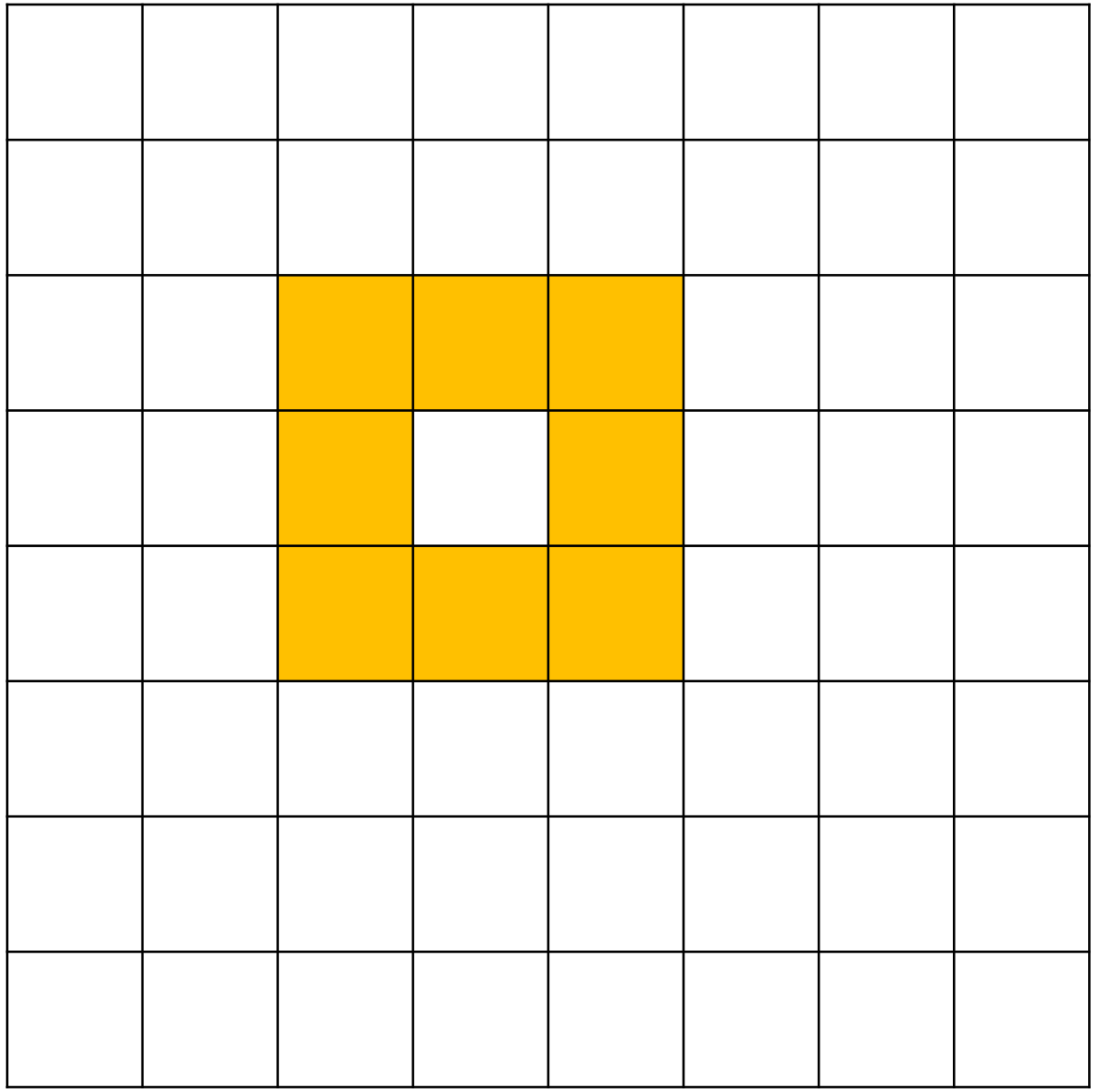
2. Count live neighbors



3. Update for next generation

		1	1	1			
1	2	3	2	3	2	1	
1	1	3	4	3	1	1	
1	2	3	2	3	2	1	
		1	1	1			

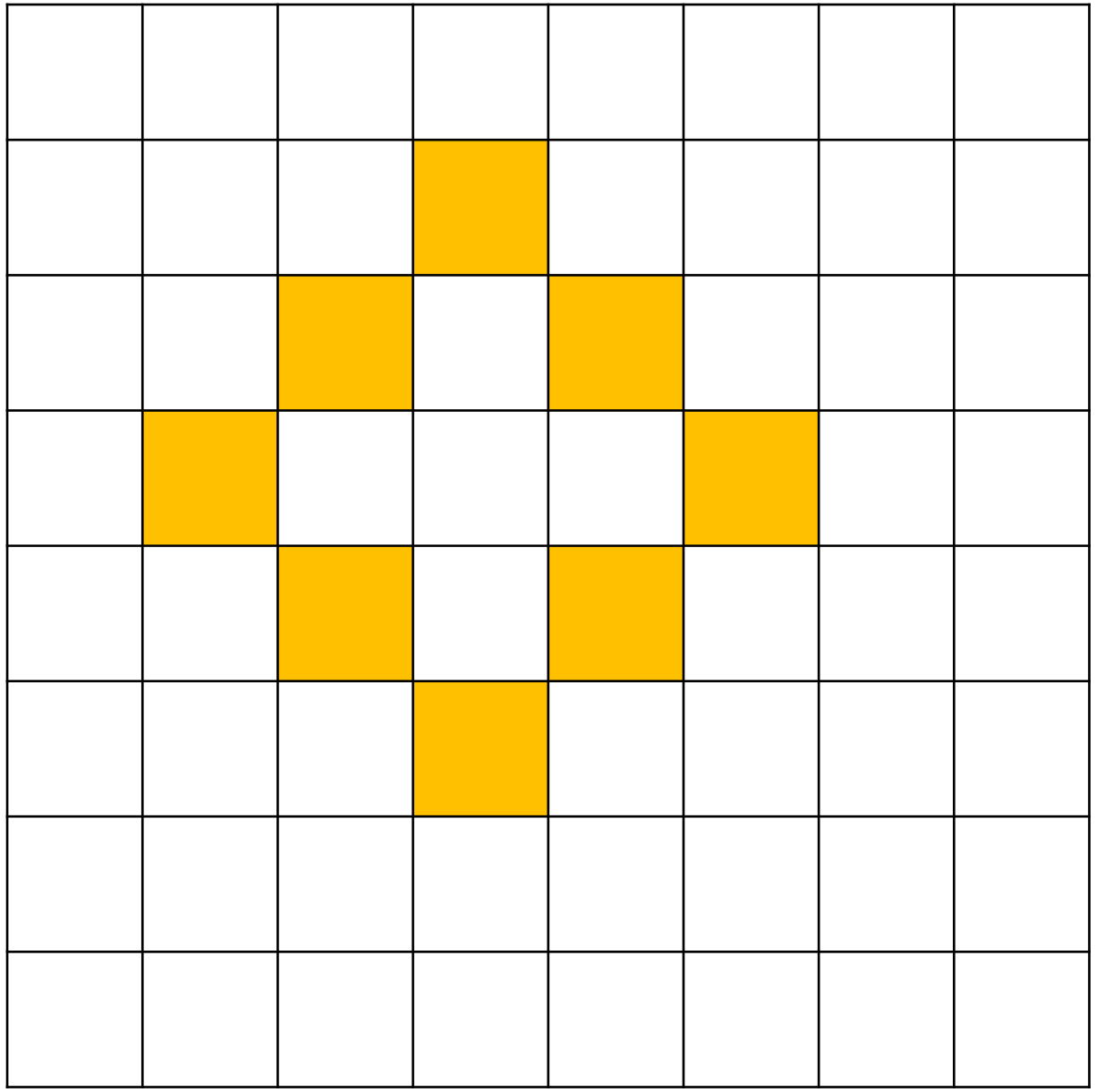
2. Count live neighbors



3. Update for next generation

	1	2	3	2	1		
	2	2	4	2	2		
	3	4	8	4	3		
	2	2	4	2	2		
	1	2	3	2	1		

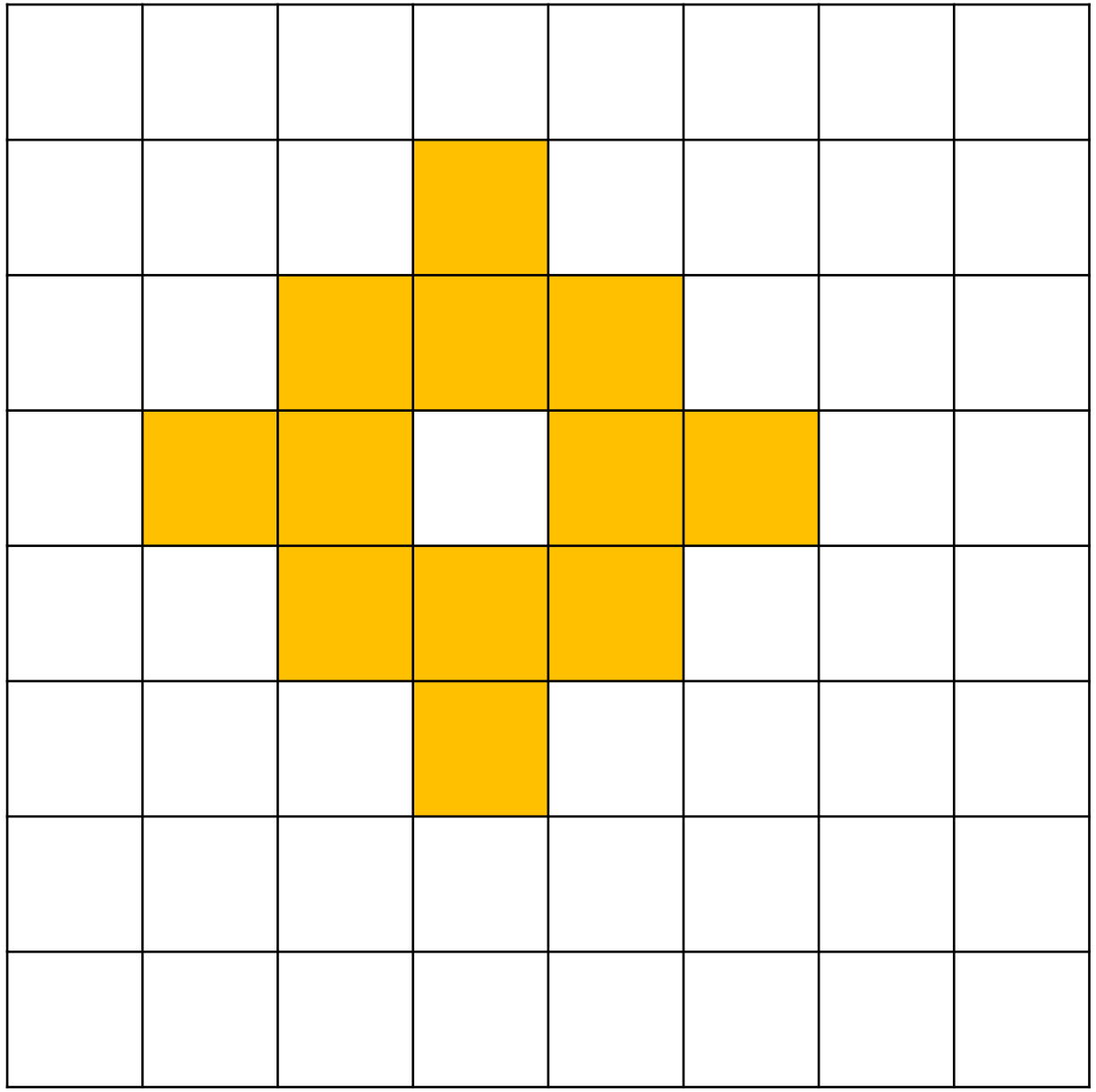
2. Count live neighbors



3. Update for next generation

		1	1	1			
	1	2	2	2	1		
1	2	2	3	2	2	1	
1	2	3	4	3	2	1	
1	2	2	3	2	2	1	
	1	2	2	2	1		
		1	1	1			

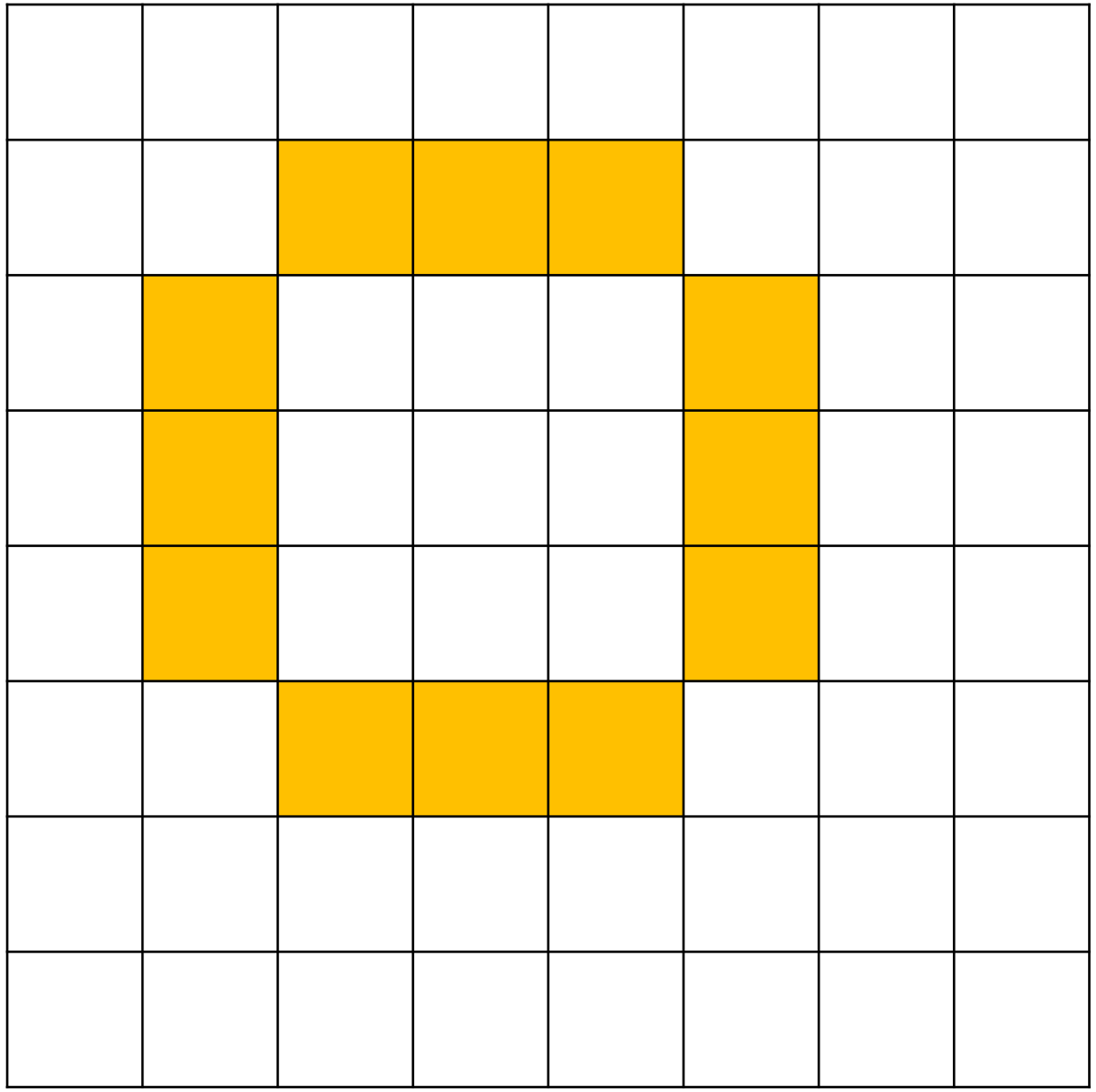
2. Count live neighbors



3. Update for next generation

		1	1	1			
	1	3	3	3	1		
1	3	4	5	4	3	1	
1	3	5	8	5	3	1	
1	3	4	5	4	3	1	
	1	3	3	3	1		
		1	1	1			

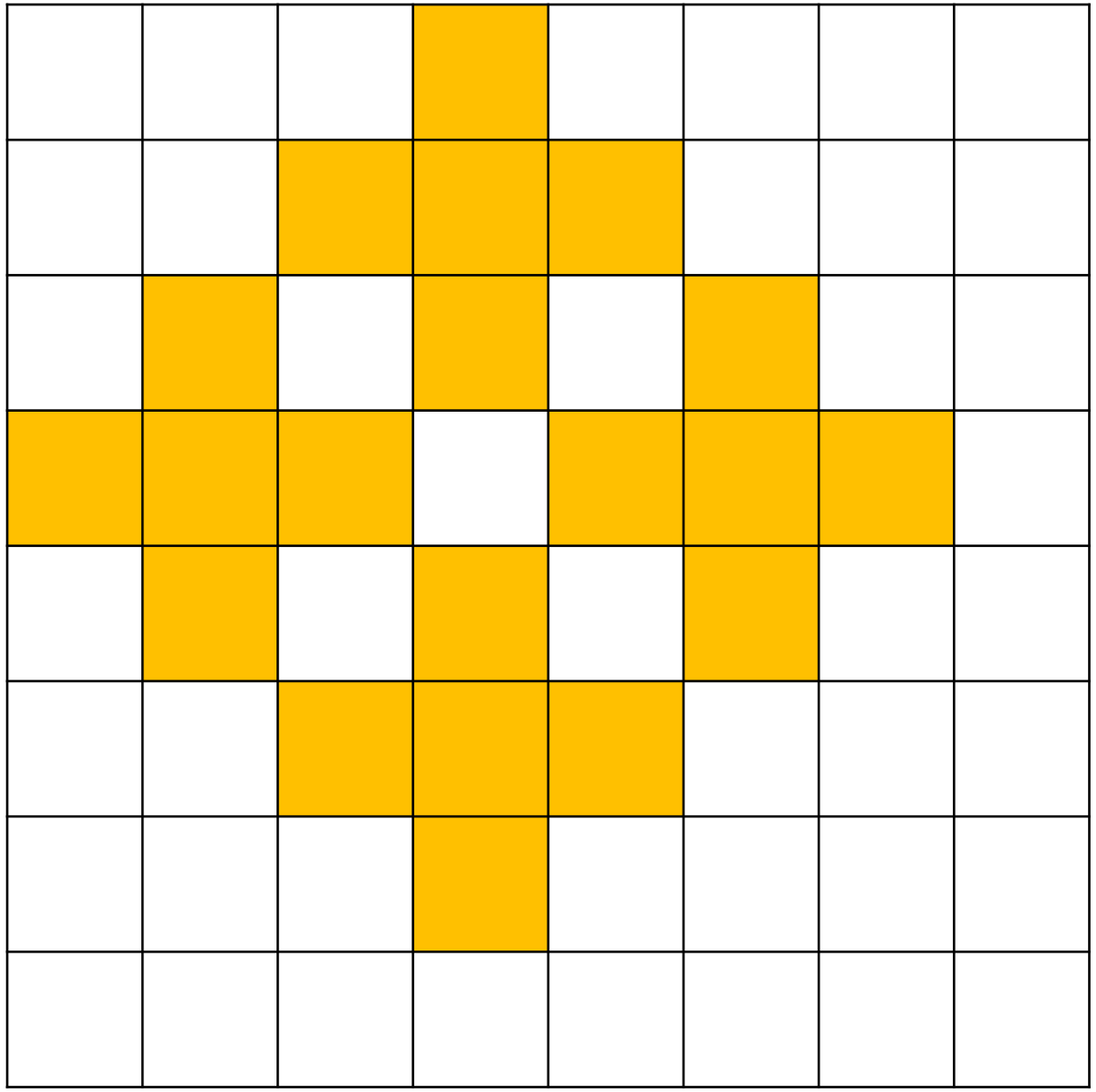
2. Count live neighbors



3. Update for next generation

	1	2	3	2	1		
1	2	2	2	2	2	1	
2	2	4	3	4	2	2	
3	2	3		3	2	3	
2	2	4	3	4	2	2	
1	2	2	2	2	2	1	
	1	2	3	2	1		

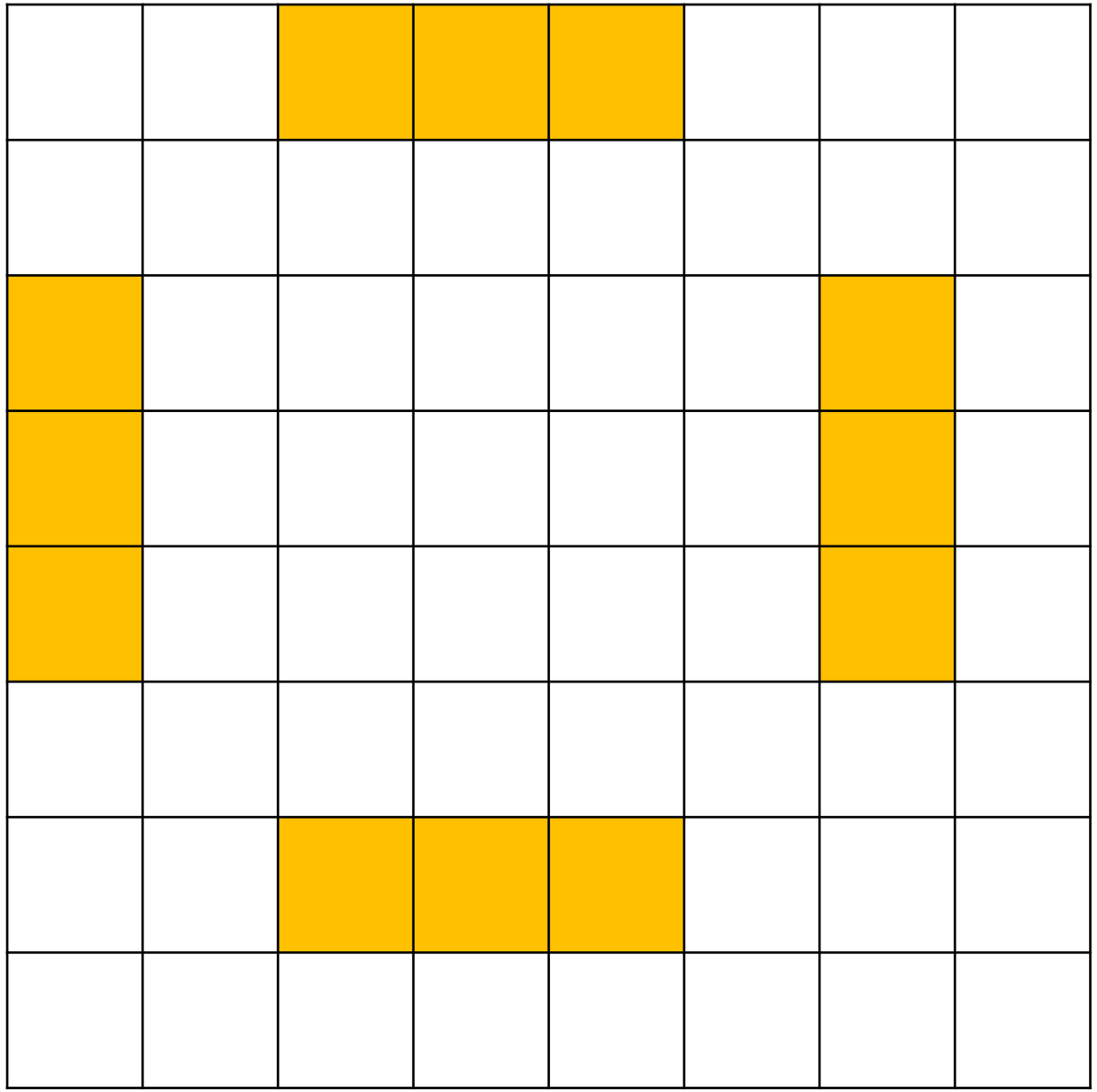
2. Count live neighbors



3. Update for next generation

	1	3	3	3	1		
1	2	4	4	4	2	1	
3	4	4	5	4	4	3	1
3	4	5	4	5	4	3	1
3	4	4	5	4	4	3	1
1	2	4	4	4	2		
	1	3	3	3	1		
		1	1	1			

2. Count live neighbors



3. Update for next generation

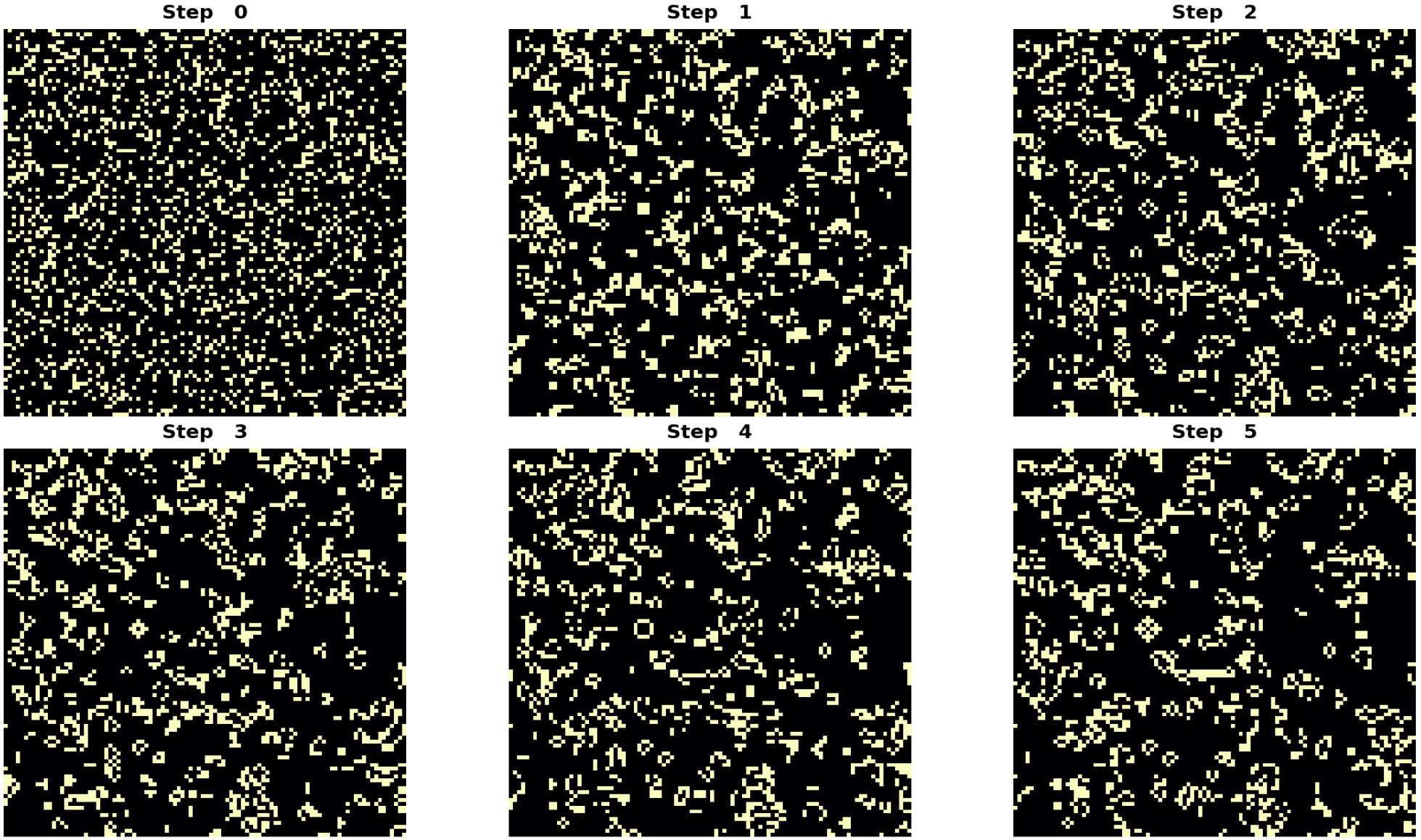
	1	1	2	1	1		
1	2	2	3	2	2	1	1
1	2				2	1	2
2	3				3	2	3
1	2				2	1	2
1	2	2	3	2	2	1	1
	1	1	2	1	1		
	1	2	3	2	1		

2. Count live neighbors

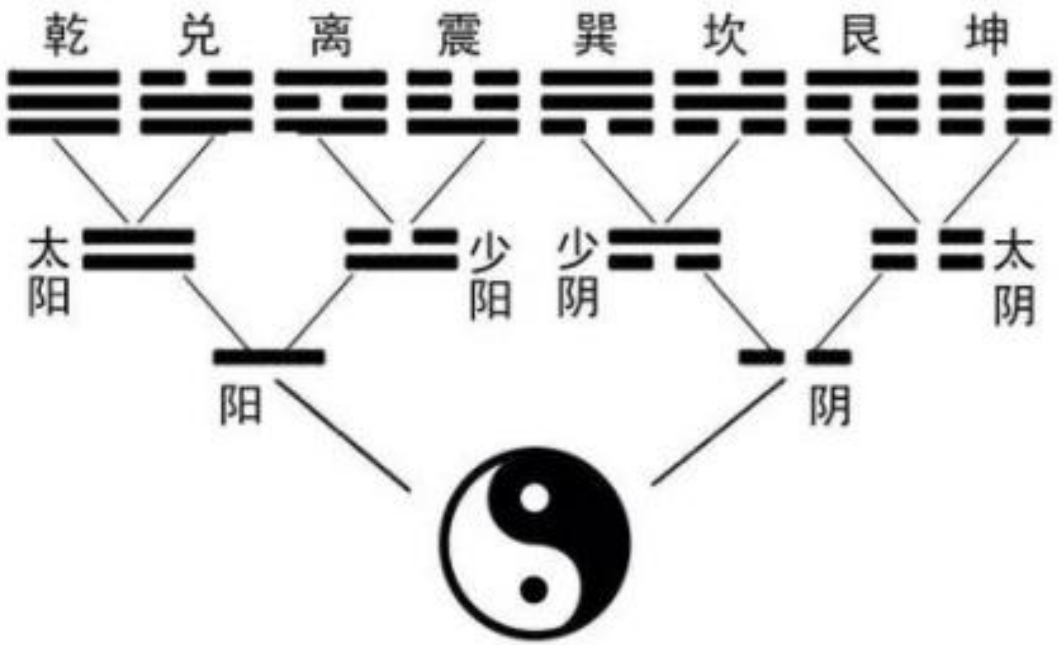
	1	1	2	1	1		
1	2	2	3	2	2	1	1
1	2				2	1	2
2	3				3	2	3
1	2				2	1	2
1	2	2	3	2	2	1	1
	1	1	2	1	1		
	1	2	3	2	1		

3. Update for next generation

2D Cellular Automaton: Conway's Game of Life
(100×100 Grid, Periodic Boundaries)



“道生一，一生二，二生三，三生万物”
“The Tao produced One; One produced Two; Two produced Three; Three produced all things.”

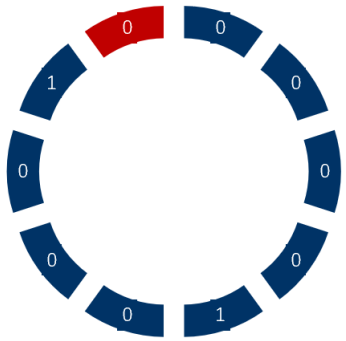


Given world

0	0	0	0	0	1	0	0	0	1

How about borders:

1. Continuous border:



Obtain neighbors

10**0**, 000, 000, 000, 001, 010, 100, 000, 001, 01**0**

“无为而治”

“Governing by non-action.”

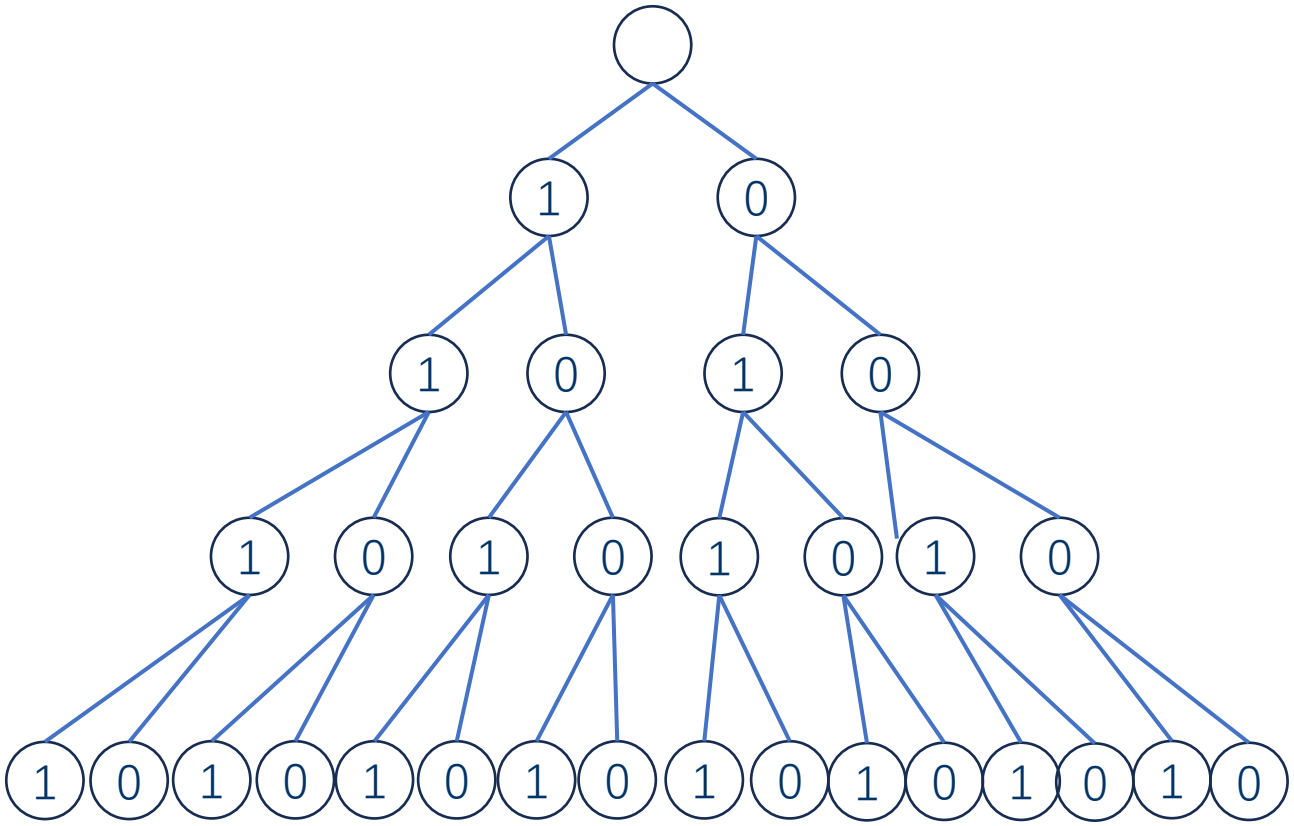
1	2	3	4	5

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15

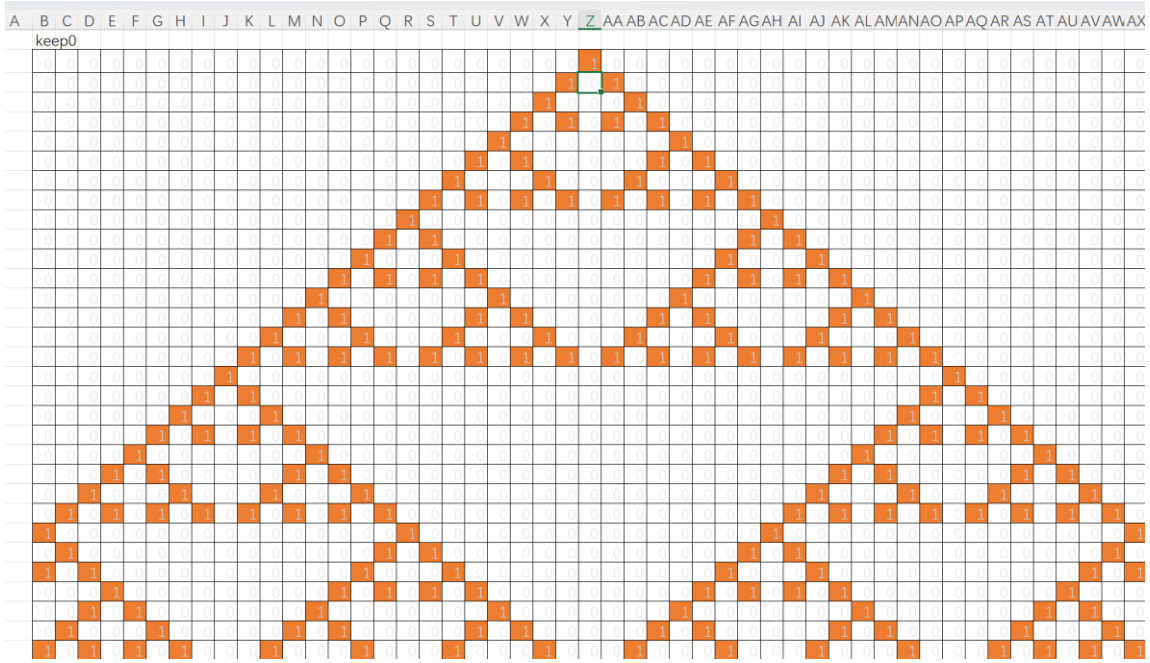
More neighbors

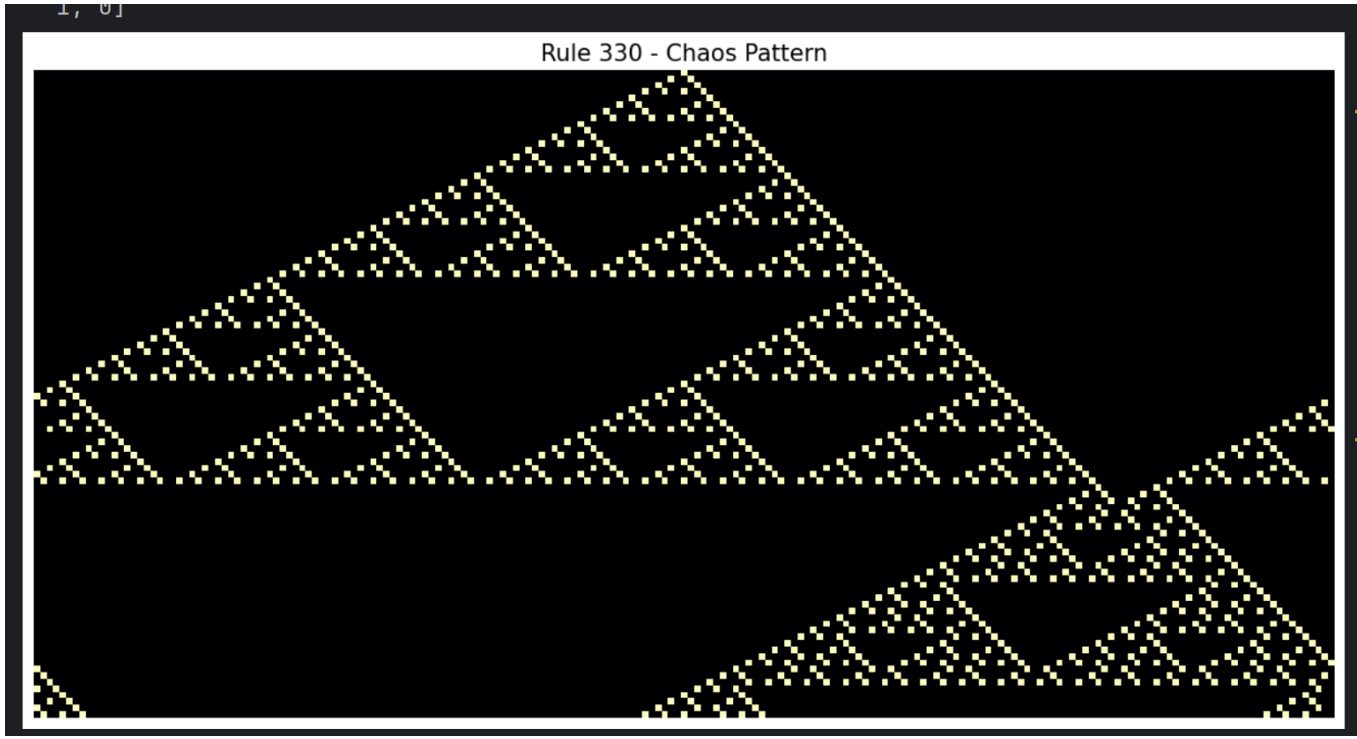
“大音希声，大象无形”

“The greatest music has the faintest notes; the greatest icon is without form.



Dec	2^{15}	2^{14}	2^{13}	2^{12}	2^{11}	2^{10}	2^{9}	2^{8}	$2^{7}=128$	$2^{6}=64$	$2^{5}=32$	$2^{4}=16$	$2^{3}=8$	$2^{2}=4$	$2^{1}=2$	$2^{0}=1$
Bin	1111	1110	1101	1100	1011	1010	1001	1000	0111	0110	0101	0100	0011	0010	0001	0000


keep0


Rule 330 - Chaos Pattern